

DOHYO JEONG

Ph.D.

Department of Internal Medicine

University of Kentucky

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RESEARCH INTERESTS

Research Topic: Public Health, Policy Spillover Effects, Community and Health Disparities, Regionalization

Methodology Topic: Machine Learning, Geographic Information Science (GIS), Spatial Econometrics

EMPLOYMENT

2025 to present **Postdoctoral Scholar,**

Department of Internal Medicine, University of Kentucky

- Building AI-driven spatial models to evaluate regional diffusion of public health interventions and guide equitable policy design.
- Analyzing opioid overdose response systems using spatial and causal methods to assess intervention effectiveness and treatment access.

EDUCATION

2020 to 2025 **The University of Texas at Dallas, Texas, USA**

Ph.D. in Public Policy and Political Economy (Spring 2025).

Dissertation Title: Regional Disparities in Global Public Health: Crisis Management, Resource Allocation, and Social Impact (Committee Chair: Dr. Dohyeong Kim)

2024 to 2025 **The University of Texas at Dallas, Texas, USA**

Master of Geospatial Information Sciences (Spring 2025)

Thesis Title: Volatility in Region, Variability in Prediction: Spatial Drivers of Public Health Fiscal Errors in South Korea (Advisor: Dr. Dohyeong Kim)

2017 to 2019 **SUNGKYUNKWAN UNIVERSITY, SEOUL, KOREA**

Master of Public Administration (GPA: 4.45/4.5)

Thesis Title: Factor Affecting the Distribution of National Subsidies in Korean Local Governments: Focusing on Rhodes' Power-Dependence Model (Advisor: Dr. SUHO BAE)

2011 to 2017 **SEOKYEONG UNIVERSITY, SEOUL, KOREA**

Bachelor of Arts in Philosophy & Public Administration (GPA: 4.34/4.5)

PUBLICATIONS

[5] **DOHYO JEONG**, Hanson-DeFusco, J., Kim, D., & Lee, C. K. (2024). Digital Mass Hysteria during Pandemic? A Study of Twitter Communication Patterns in the US during the Stages of COVID-19 Vaccination. *Behavioral Sciences*, 14(5), 389.

<https://doi.org/10.3390/bs14050389> (IF: 2.6)

- Research Topic: Analysis of changes in public sentiment regarding vaccine supply policy
- Design/Method: Text Sentiment Analysis and Interrupted Time Series Analysis.

[4] Hong, S., **DOHYO JEONG**, & Kim, P. (2024). Have offender demographics changed since the COVID-19 Pandemic? Evidence from money mules in South Korea. *Journal of Criminal Justice*, 91, 102156. <https://doi.org/10.1016/j.jcrimjus.2024.102156> (IF: 5.5)

- Research Topic: Exploring the Influence of the COVID-19 Pandemic on Crime Patterns
- Design/Method: Interrupted Time Series Analysis.

[3] **DOHYO JEONG.**, Kim, D., Mohiuddin, H., Kang, S., & Kim, S. (2023). Regional Disparity in the Educational Impact of COVID-19: A Spatial Difference-in-Difference Approach. *Sustainability*, 15(16), 12514. <https://doi.org/10.3390/su151612514> (IF: 3.9)

- Research Topic: Analyzing regional disparities in transitioning to online classes.
- Design/Method: Spatial Difference-in-Difference.

[2] Odusola, A. O., **DOHYO JEONG**, Malolan, C., Kim, D., Venkatraman, C., Kola-Korolo, O., ... & Nwariaku, F. E. (2023). Spatial and temporal analysis of road traffic crashes and ambulance responses in Lagos state, Nigeria. *BMC public health*, 23(1), 2273. <https://doi.org/10.1186/s12889-023-16996-8> (IF: 4.7)

- Research Topic: Comparing traffic accident patterns based on time, and regional characteristics.
- Design/Method: Geospatial mapping and Hotspot Analysis.

[1] Ogbudebe Chidubem, **DOHYO JEONG**, OdumeBethrand (2023). Editorial Decision/Comments on "Identifying Hotspots of Tuberculosis in Nigeria using Early Warning Outbreak Recognition System: Retrospective Analysis of Implications for Active Case Finding Interventions. *JMIR Public Health and Surveillance*, 9 (1), e40311.

<https://publichealth.jmir.org/2023/1/e40311/> (IF: 8.5)

- Research Topic: Evaluation of the effectiveness of tuberculosis early warning program
- Design/Method: Kernel density and Getis-Ord Gi* Hot Spot Analyses.

Under Review paper

[5] **DOHYO JEONG**, Daniel Harris. (2025). Emerging Bromazepam-Related Overdose Deaths: Spatial Hotspots and Social Determinants.

- Research Topic: Environmental exposure impacts on opioid-related overdose deaths.
- Design/Method: Propensity score matching and spatial regression discontinuity design.

[4] **DOHYO JEONG**, Dohyeong Kim, Sunghwan Cho. (2025). Where and When? Predicting Emergency Medical Service Demands through Spatial-Temporal Graph Convolutional Networks.

- Research Topic: Identification of medical resource imbalances and optimal allocation.
- Design/Method: ST-Graph Convolutional Network and Maximal Covering Location Problem.

[3] **DOHYO JEONG**, Dohyeong Kim, Chang-jin Kim. (2025). Fiscal Forecast Errors in Public Health Expenditure: Based on a Spatial-Temporal Approach.

- Research Topic: Identifying medical resource supply and demand imbalance and determining factors
- Design/Method: Calculating Regional Variability, Spatial Regression Model.

[2] **DOHYO JEONG**, Dohyeong Kim, Okey Okuzu, Chidubem Ogbudebe. (2025). Beyond Distance: Does Geographical Accessibility for Tuberculosis Treatment Follow Supply, Demand or Both?

- Research Topic: The impact of geographical accessibility on tuberculosis treatment initiation
- Design/Method: Calculating the regional geographic accessibility, Generalized Linear Mixed Models

[1] **DOHYO JEONG**, Dohyeong Kim (2025). Bridging Gaps or Widening Divides? A Spatiotemporal Analysis of COVID-19 Therapeutic Distribution Policies in Texas

- Research Topic: Identifying medical resource supply and demand imbalance and determining factors
- Design/Method: Spatial-Temporal Integrated Nested Laplace Approximations (INLA).

Publications in Korea

[6] CHANG-JIN KIM, **DOHYO JEONG**, (2022). Factors Influencing Public Officials Innovative Behavior for Platform Governance. *The Journal of Korea Policy Research*. Vol.22 No.3: 141-171.

- Research Topic: Analyzing innovation factors in public sector platform governance.
- Design/Method: Platform governance and Factors influencing innovative behavior.

[5] **DOHYO JEONG**, Sangho Moon, SUHO BAE. (2019). Factors Affecting the Distribution of National Subsidies in Korean Local Governments: Focusing on Rhodes' Power-Dependence Model. *The Korea Journal of Policy Analysis and Management*, Vol.29 No3: 21-53.

- Research Topic: Factors in central subsidy allocation to local governments
- Design/Method: Panel Corrected Standard Errors (PCSE) model and Prais-Winsten procedure.

[4] **DOHYO JEONG**, CHANG-JIN KIM, SUHO BAE. (2019). A Study on Determinants of Tax Attitude: Focusing on Slippery Slope Framework, *Korea Public Policy Review*, Vol.33 No.3: 43-72.

- Research Topic: Examining the Impact of Tax Compliance and Taxpayer Attitude
- Design/Method: Decision tree model and neural network model.

[3] CHANG-JIN KIM, **DOHYO JEONG**, SUNG-WOO HONG. (2019). The Effects of Decentralization Perception on the Recognition of Intergovernmental Relationship: Focusing on Mediating Effect of Dispute Settlement System - *Journal of Local Government Studies*. Vol.31 No.3: 1-35.

- Research Topic: Government Perception and Decentralization
- Design/Method: Ordered Logit Model and Structural Equation Model (SEM).

[2] **DOHYO JEONG**, YOUNGKYU LEE, SEONGYOUNG JEONG. (2018). An Analysis of the Effect of the Tax Rate on the Financial Efficiency of Local Governments. *The Korea Journal of Local Government Studies*, Vol.22 No.3: 415-443.

- Research Topic: Local Tax Flexibility and Fiscal Efficiency
- Design/Method: Difference-in-Difference analysis, Panel Corrected Standard Errors model.

[1] Dae-yong Hyun, **DOHYO JEONG**, (2017). Analysis of differences in perception of administrative values among civil servants and general civil servants: Focused on Suwon City Government Officials. *Suwon Research Institute*. No. 12: 119-141.

- Research Topic: Comparing administrative values between City officials and general civil servants.
- Design/Method: Ordered Logit Model.

CONFERENCE PRESENTATION

[9] **DOHYO JEONG**. (2025). Predicting Emergency Medical Service Demands through Spatial-Temporal Graph Convolutional Networks. ASPA (American Society for Public Administration) 2025 Annual Conference. March. 2025.

[8] **DOHYO JEONG**. (2024). Beyond Distance: Does Geographical Accessibility for Tuberculosis Treatment Follow Supply, Demand or Both? APHA (American Public Health Association) 2024 Annual Conference. October. 2024.

[7] **DOHYO JEONG**. (2024). Fiscal Forecast Errors in Public Health Expenditure: Based on a Spatial-Temporal Approach. ASPA (American Society for Public Administration) 2024 Annual Conference. April. 2024

[6] **DOHYO JEONG**. (2023). Differential Side Effects of COVID-19 Response Policies on the U.S. Labor Market: A Spatial-Temporal Analysis. APPAM (Association for Public Policy Analysis and Management) 2023 Annual Conference. Nov. 2023.

[5] **DOHYO JEONG**. (2023). The Patterns of COVID-19 Therapeutics Supply and Demand in Texas: A Spatial-Temporal INLA Approach. APHA (American Public Health Association) 2023 Annual Conference. Nov. 2023.

[4] **DOHYO JEONG**. (2023). Regional disparity in the uninsurance rate impact of COVID-19: a spatial machine learning approach. ASPA (American Society for Public Administration) 2023 Annual Conference. 21. March. 2023.

[3] **DOHYO JEONG**. (2023). Digital Mass Hysteria? during Pandemics: A Case Study of Twitter Communication Patterns in the US during COVID-19 Period. Conference On Public Process Research. 12. Jan. 2023.

[2] **DOHYO JEONG**. (2022). A comparison of the spread trend prediction model according to the government's COVID-19 response policy change and its influence, 2022 Korean Public Administration International Conference. Korea. 22 June. 2022.

[1] **DOHYO JEONG**. (2021). The effect of the government's vaccination management plan on the change of sentiment toward vaccines, 2021 Global Disastronomy Workshop. Texas. USA. 17 Dec. 2021.

Presentation in Korea

[5] **DOHYO JEONG**, Chang-jin Kim. SUHO BAE. (2019). A Study on the Factors Affecting the Taxation Attitude of General Taxpayers. Korean Association for Local Government Studies Winter Conference. Seoul. KOREA. 14 Feb. 2019

[4] Chang-jin Kim. **DOHYO JEONG**, SUHO BAE. (2019). The Mediation Effect of Dispute Settlement System in the Perception of Regional Dispersion and Intergovernmental Relations. Korean Association for Local Government Studies Winter Conference. Seoul. KOREA. 14 Feb. 2019.

[3] **DOHYO JEONG**, YOUNGKYU LEE, SUHO BAE. (2018) An Analysis of the Effect of the Tax Rate on the Financial Efficiency of Local Governments. Korea Association of Local Administration Summer Joint Conference Chungcheong-do. KOREA. 20 Jul. 2018

[2] **DOHYO JEONG**. (2017). The Effects of Tax Recognition on the pros and cons of Welfare Policy. Seoul Association of Public Administration Fall Conference. Seoul. KOREA. 3 Nov. 2017

[1] Dae-yong Hyun, **DOHYO JEONG**. (2017). A Study on the Policy Diffusion of Local Government in Korea: focusing on Resident Participation Budget System. Korea Association of Local Administration Summer Joint Conference. Gyeonggi-do. KOREA. 18 Aug. 2017

AWARDS, SCHOLARSHIPS AND HONORS

In the US

Apr 2025	Outstanding Graduate Student Award (PhD) in The University of Texas at Dallas
March 2025	1st prize in Student Paper Award Issued by Section on Korean Public Administration of American Society for Public Administration (ASPA)
May 2024	Dean of Graduate Education Dissertation Research Award in The University of Texas at Dallas
Apr 2024	2 nd prize in Student Paper Award Issued by Section on Korean Public Administration of American Society for Public Administration (ASPA)
Fall 2023	President Travel Award for Association for Public Policy Analysis and Management (APPAM), The Korean Association for Policy Studies

- Spring 2023 Awarded Government and Political Science Fellowship in The University of Texas at Dallas
- Spring, 2023 Nominated for President's Teaching Excellence Awards in The University of Texas at Dallas

In Korea

- Sep. 2022 Awards 3rd prize in the treatise contest by the Korea Institute of Public Finance
- Feb. 2019 Awards for Excellent Records, Sungkyunkwan University, South Korea
- Nov. 2018 Awards The 3rd Report for Korea National Tax Administration Policy Proposal Contest Participation Prize, National Tax Service
- Fall 2018 Academic Excellence Scholarship, Sungkyunkwan University, South Korea
- Spring 2018 Academic Excellence Scholarship, Sungkyunkwan University, South Korea
- Fall 2017 Academic Excellence Scholarship, Sungkyunkwan University, South Korea
- Spring 2017 Encouragement of Research Scholarship, Sungkyunkwan University, South Korea
- Fall 2016 An Excellent-grade Scholarship, Seokyeong University, South Korea
- Spring 2016 An Excellent-grade Scholarship, Seokyeong University, South Korea
- Fall 2015 Grade(extra) Scholarship, Seokyeong University, South Korea
An Excellent-grade Scholarship, Seokyeong University, South Korea
- Spring 2015 Recommendation of Department Scholarship, Seokyeong University, South Korea
- Fall 2014 Special Scholarship, Seokyeong University, South Korea
- Fall 2011 An Excellent-grade Scholarship, Seokyeong University, South Korea

RESEARCH EXPERIENCE

May. 2025 - Rapid Actionable Data for Opioid Response in Kentucky (RADOR-KY)

- Postdoctoral Scholar, University of Kentucky

- Used machine learning and spatial analysis to model EMS responses to opioid overdoses.
- Identified region-level disparities in emergency response capacity and delays.
- Integrated EMS and socioeconomic data to evaluate community vulnerability and risk.
- Generated evidence to support data-informed decisions on emergency resource allocation.

May. 2025 - Geospatial Pipeline for Enhancing Open Data for Substance Use Disorders Research (geoPIPE)

- Postdoctoral Scholar, University of Kentucky

- Developed an AI-supported geospatial pipeline to improve open data for substance use policy research.
- Applied spatial clustering and predictive modeling to identify fatal overdose hotspots.
- Analyzed relationships among overdose clusters, social determinants, and policy environments.
- Produced tools that enhance data accessibility and strengthen evidence-based public health policy design.

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- Feb. to Dec. 2023 **Medical Resource Optimization Model Analysis for Vulnerable Areas of Emergency Medical Service Using Machine Learning: A Case Study on Korea.**
- Research Assistant, University of Texas at Dallas
- Traffic accident case data collection and mapping.
 - Demand-supply imbalance analysis for medical resources.
 - Optimization model calculation through machine learning.
 - Policy proposals and reports for optimizing medical resources.
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- Dec. 2021 to June. 2022 **Identifying Hotspots of Tuberculosis in Nigeria using EWORS: Implications for Active Case Finding and Intervention.**
- Research Assistant, University of Texas at Dallas
- Combining and managing Nigerian tuberculosis data
 - Disease distribution mapping and pattern analysis
 - EWORS system effectiveness analysis and evaluation
 - System effect mapping and geospatial analysis
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- Aug. 2021 to Dec. 2022 **Emergency Medical Services (EMS) Policy Implementation Evaluation and Efficiency Analysis in Nigeria**
- Research Assistant, University of Texas at Dallas
- Nigeria traffic accident information mapping and data management
 - Ambulance Reaction Time Patterns and Statistical Analysis
 - Analysis and evaluation of traffic policy effects
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- Mar. 2017 to Feb. 2019 **Pursuing an Empathic Government Through Convergent Leadership Development Program**
- Research Assistant, SUNGKYUNKWAN UNIVERSITY/NATIONAL RESEARCH FOUNDATION OF KOREA
- Party of support: National Research Foundation of Korea
 - Responsibilities:
 1. Diagnosis and development of problems in interdisciplinary convergence research
 - Compare the concept of ‘Publicity’ in the West and the East, and study the definition of publicity in Korean Public Administration
 - Compare the bureaucracy theory of the Oriental administration Philosopher Han Feizi and the bureaucracy theory of Max Weber in the West, and derive various implications for application to modern bureaucracy
 2. Analysis and evaluation of the influence of government trust
- Study the influence of tax recognition and government trust on the welfare policy attitude

TEACHING EXPERIENCE

- 2024 Fall EPPS Math and Coding Camp
- The University of Texas at Dallas
 - In-person Class
 - The EPPS Math and Coding Camp is designed to provide first-year EPPS graduate students with a week of training in fundamental mathematics and basic programming. The camp focuses on equipping and inspiring students with applicable math and data analysis techniques for their upcoming graduate research methods courses. I managed the program in this camp and taught students coding, primarily focusing on R.
 - [EPPS Math and Coding Camp homepage](#)
- 2024 Summer GIS and Spatial Statistics for Social Science: Advanced
- Korea Social Science Data Archive (KOSSDA) at Seoul National University
 - Online Class
 - This course is for students or researchers with basic experience in spatial data analysis or GIS. It provides hands-on training in advanced spatial statistical methods, including 2SFCA, GLMM, and spatiotemporal Bayesian models, using R.
 - Instructor Evaluation: 9.78/10
- 2024 Summer GIS and Spatial Statistics for Social Science: Basic
- Korea Social Science Data Archive (KOSSDA) at Seoul National University
 - Online Class
 - This course introduces key GIS techniques such as mapping, buffer analysis, nearest-neighbor analysis, and spatial regression. Students learn to analyze spatial datasets, extract meaningful patterns, and practice core spatial methods using real social science data.
 - Instructor Evaluation: 9.81/10
- 2024 Winter Big Data Analysis and Machine Learning for Social Science
- Korea Social Science Data Archive (KOSSDA) at Seoul National University
 - Online Class
 - This course aims to establish theoretical foundations to understand how machine learning techniques, actively developed and applied in recent years in science and engineering, can be utilized to analyze big data constructed in the social sciences. After grasping the basic structures of representative supervised and unsupervised learning methods, our main objective is to conduct practical exercises using R packages with various examples.
 - Instructor Evaluation: 9.70/10

2023 Fall EPPS 2302. Methods of Quantitative Analysis in the Social and Policy Sciences (Fall, 2023)

- The University of Texas at Dallas
- In-person Class
- This course introduces basic concepts and methods of statistical analysis used in different fields of social and policy science research to better understand human relationships and the impacts of government action on them. Topics include data description, using probability to assess the reasonableness of claims about the world based on sample data, exploring cause-effect interactions through regression models, and application of software to ease visualization and calculation. Students completing this course will be good consumers of statistical information and have a solid foundation for pursuing further study of quantitative analysis.
- [Syllabus](#)
- Instructor Evaluation: 4.75/5

GUEST LECTURES

2024 Spring SOC 4385 Global Health and Society (Spring, 2024)

- The University of Texas at Dallas
- In this guest lecture for the Global Health and Society course, I introduced the concepts of spatial statistics and mapping as they apply to the social sciences. The lecture was designed to provide theoretical knowledge and practical skills, focusing on the use of GeoDa software. This lecture began with an overview of spatial statistics, discussing its importance and relevance in analyzing geographic and demographic data. Following the theoretical introduction, we transitioned to a hands-on session using GeoDa, a user-friendly spatial data analysis software. This lecture explored how to create various types of maps, such as choropleth maps, to visualize data distributions and identify spatial patterns. The practical session included exercises on how to apply spatial statistical methods to real-world data, allowing students to gain a deeper understanding of the techniques discussed earlier.

TEACHING ASSISTANTS

2024 Fall EPPS 2302. Methods of Quantitative Analysis in the Social and Policy Sciences
Instructor: Dr. Timothy Bray

2024 Spring EPPS 7316. Regression and Multivariate Analysis
Instructor: Dr. Patrick Brandt

2023 summer	Advanced GIS and Spatial Statistics • Korea Social Science Data Archive at Seoul National University • Rating: 9.84/10
2023 summer	Introductory GIS and Spatial Statistics • Korea Social Science Data Archive at Seoul National University • Rating: 9.54/10
2023 Spring	EPPS 7316. Regression and Multivariate Analysis Instructor: Dr. Patrick Brandt
2022 Fall	EPPS 7313. Descriptive and Inferential Statistics Instructor: Dr. Dohyeong Kim
2022 Spring	PPPE 6321. Economics for Public Policy Instructor: Dr. Dohyeong Kim

WORK EXPERIENCE

Jan. 2020 to May. 2020	Gyeonggi Research Institute, Korea <i>Researcher, Economic and Social Research Lab</i> Responsibilities 1. Local industry geographic distribution data collection and analysis. 2. Establishment and analysis of strategic plans for regional strategic industries. 3. Industry policy evaluation and report.
Apr. 2019 to JAN. 2020	Sustainable Urban Development Institute (SUDI) SUNGKYUNKWAN UNIVERSITY, SEOUL, KOREA <i>Researcher, the Center for Urban Policy Studies</i> • Responsibilities: 1. Diagnosis of problems of local autonomy and decentralization, and development of improvement measures • Assessment of local government's fiscal soundness and efficiency 2. Policy development for sustainable local government and rational creation of finance • Study on applicability of flexible taxes, tax competition, evaluation of local fiscal reconciliation system, and allocation of government subsidies 3. Database city information and utilize that information for city research and policy development data - Research and application of statistics package such as STATA, SPSS, AMOS, and SAS

VOLUNTARY/ EXTRACURRICULAR ACTIVITIES

July. 2024 - Present	UT Dallas Korean Student Association (Vice President) Managing overall organizational operations and operating student support services
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Sep. 2023 to June. 2024	UT Dallas Korean Student Association (Committee Chair) Responsible for student council member management and external cooperation communication
Aug. 2016 to Aug. 2016	Educators Without Borders, Seoul, Korea <i>Participant</i> Completion of the curriculum on cooperation in international education development
Mar. 2015 to Aug. 2015	Ministry of Foreign Affairs, Seoul, Korea <i>Supporter</i> Friends of Ministry of Foreign Affairs (MOFA): Policy promotion and write articles
Feb. 2015 to Feb. 2015	Seokyeong University, Seoul, Korea <i>A member of student monitoring team</i> Analysis and Evaluation of School Curriculum and Educational Conditions

CERTIFICATES & SKILLS

	- ArcGIS pro - PostgreSQL
Statistical Programs	- R statistical package - STATA statistical package - Python statistical package

REFERENCES

Dohyeong Kim, Ph.D.

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